

SIMATS ENGINEERING
ASSESSMENT TOOL 1: Concept Mapping

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1522

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COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions. (BL2)*

Assessment Tool Weightage: 20%

Dave's Taxonomy: Imitation (Level 1)
Question Count: 4

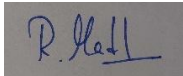
Total Marks: 50

Code of Conduct

I, Madhan R (192324256) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Assessment Tasks

Concept Mapping Tasks

Create concept maps for the following tasks. Each map must show key nodes, link labels, and a logical hierarchy.

1. Concept Map 1: Cloud Computing Fundamentals. Include definition, characteristics, benefits, and limitations.
2. Concept Map 2: Evolution of Cloud Computing. Connect hardware evolution, internet software evolution, distributed systems, and service delivery.
3. Concept Map 3: Service Models. Map IaaS, PaaS, SaaS, and XaaS with provider responsibilities and user responsibilities.
4. Concept Map 4: Virtualization and Web Services as Enablers of Cloud. Show how these two concepts support cloud deployment and scalability.

Expected concept nodes:

- Elasticity, scalability, resource pooling, on-demand access, measured service
- Virtual machine, hypervisor, abstraction, API, SOAP, REST, provider, consumer
- Infrastructure layer, platform layer, application layer, shared responsibility

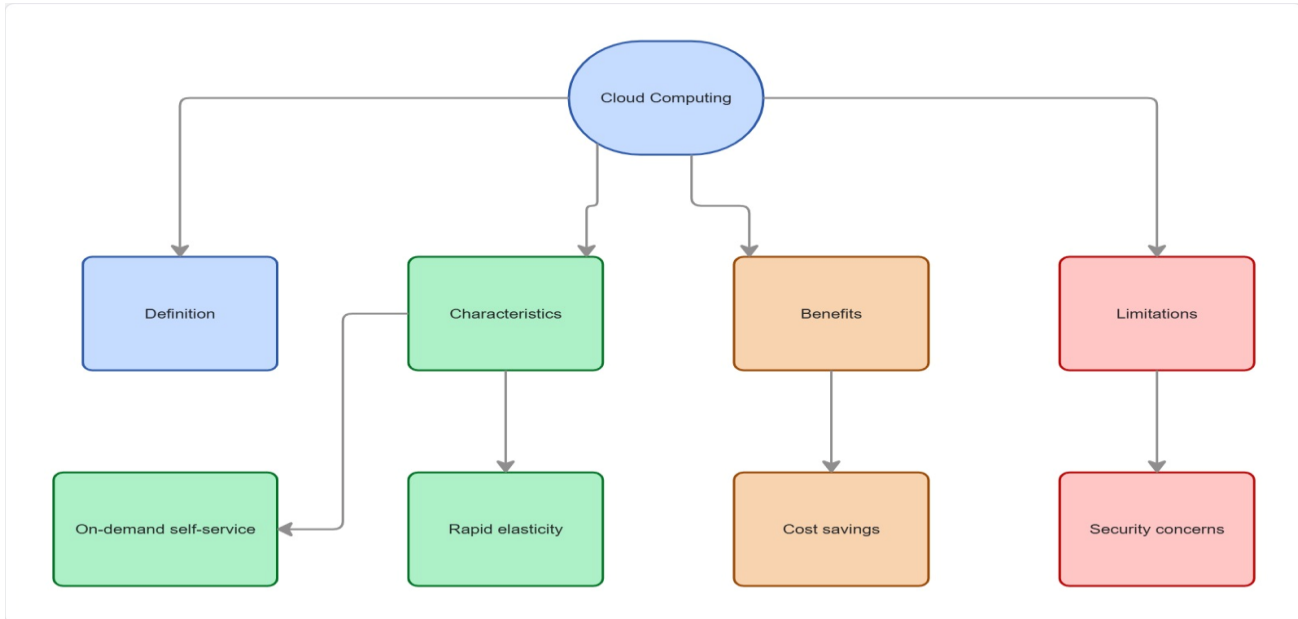
Concept Mapping Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Concept Identification	25%	All major concepts are accurately identified	Most concepts are identified correctly	Limited concepts are identified	Essential concepts are missing
Relationship Mapping	25%	Links between concepts are accurate and meaningful	Most links are correct	Some links are weak or unclear	Relationships are mostly incorrect
Hierarchy and Organization	20%	Excellent structure with clear hierarchy	Mostly organized with minor issues	Basic structure with weak hierarchy	No logical organization
Technical Relevance	20%	Map strongly reflects cloud course content	Mostly aligned to topic	Partially aligned to topic	Weak alignment to topic
Visual Clarity	10%	Neat, readable, and easy to interpret	Generally readable	Somewhat cluttered	Poorly presented

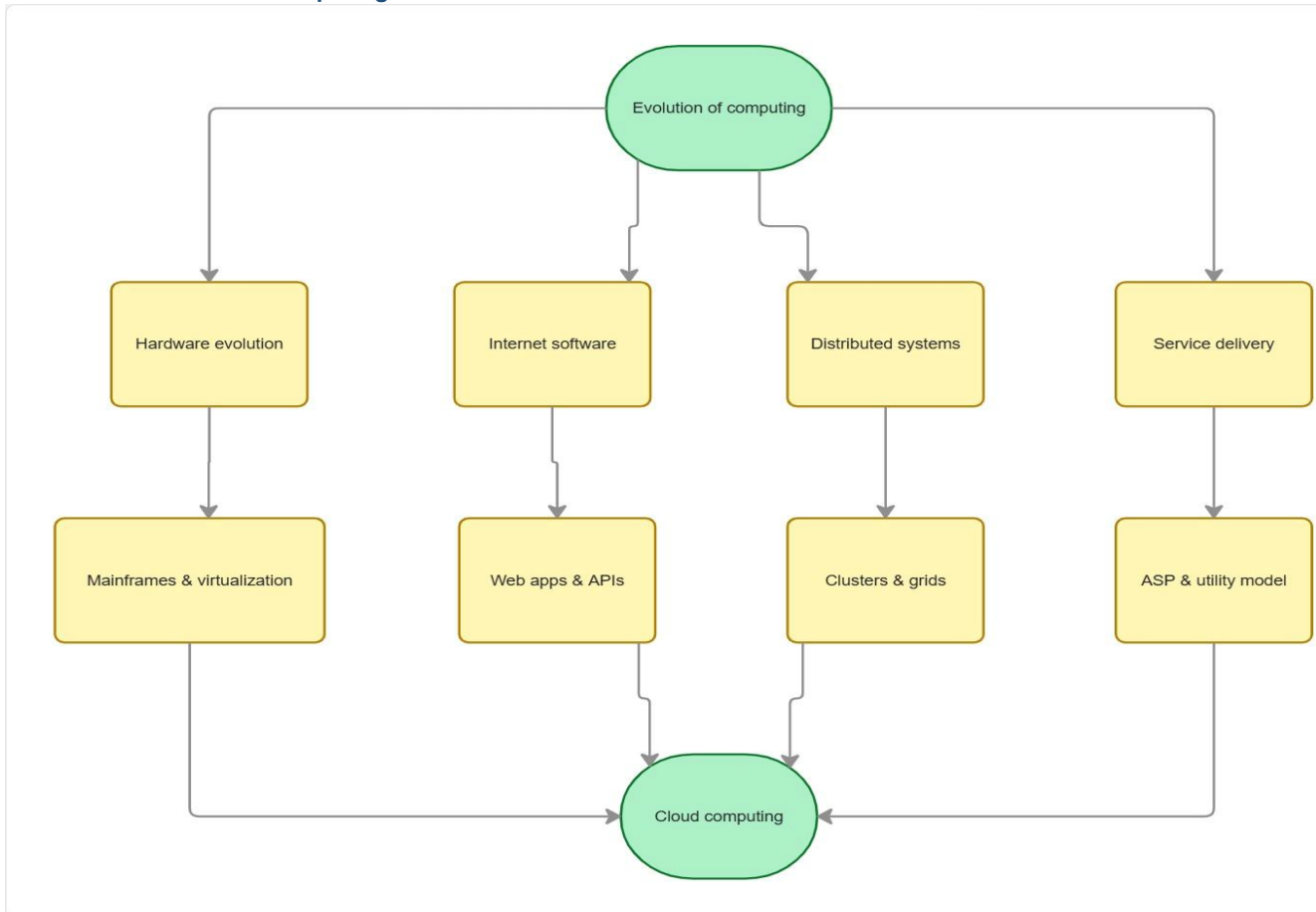
5. Questions

- i). Evaluate the effectiveness of your concept map in representing cloud computing principles. Which design decisions most strongly support your solution, and why?
- ii). Analyze how scalability, elasticity, and resource management influence the architecture shown in your concept map. What challenges could arise if these capabilities were absent?
- iii). Justify the selection of the APIs, web services, or cloud services included in your concept map. How do they contribute to the overall functionality and efficiency of the system?
- iv). Critically examine the relationships between the components in your concept map. How do these interactions improve reliability, performance, or user experience?
- v). Reflect on the process of developing your concept map. How has creating and analyzing the map changed your understanding of cloud computing architecture, and what improvements would you make if you were redesigning it?

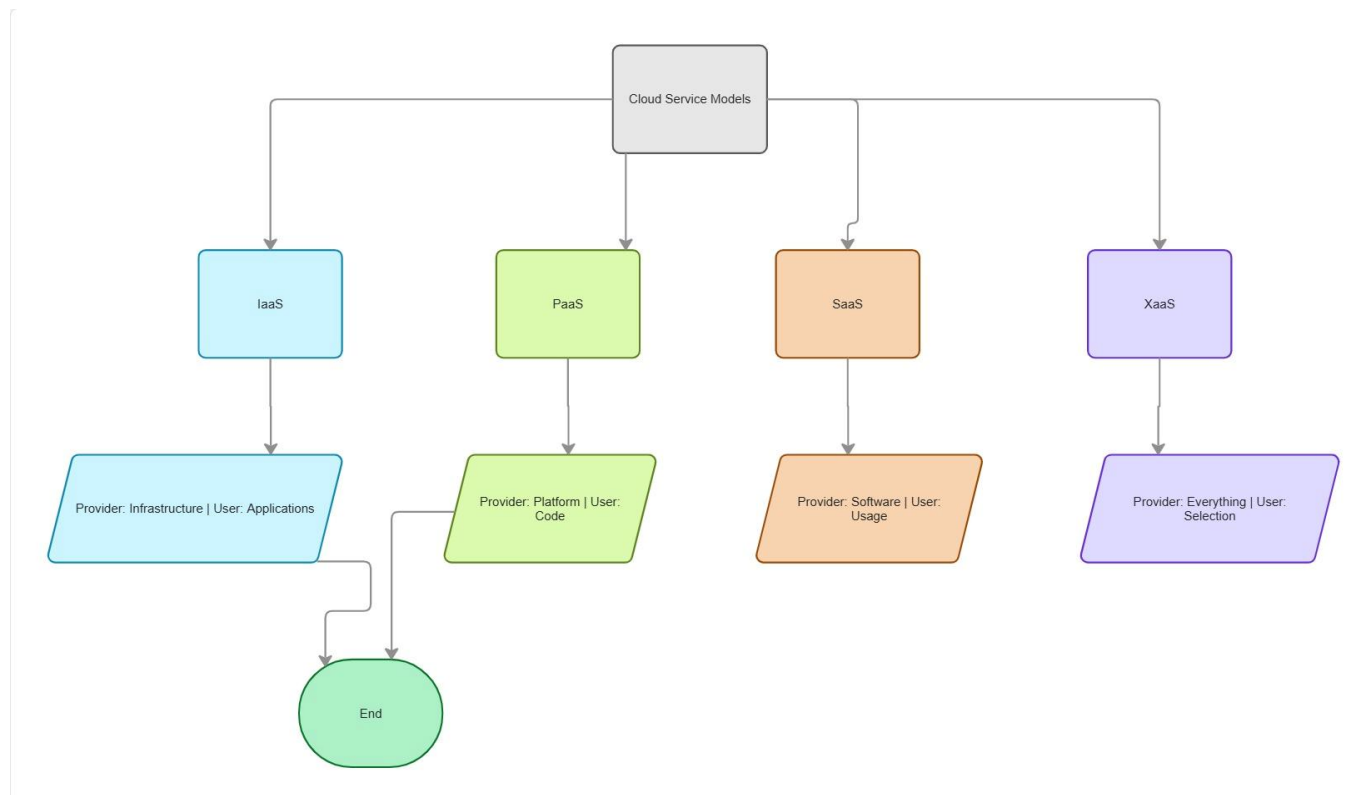
1. Cloud Computing Fundamentals



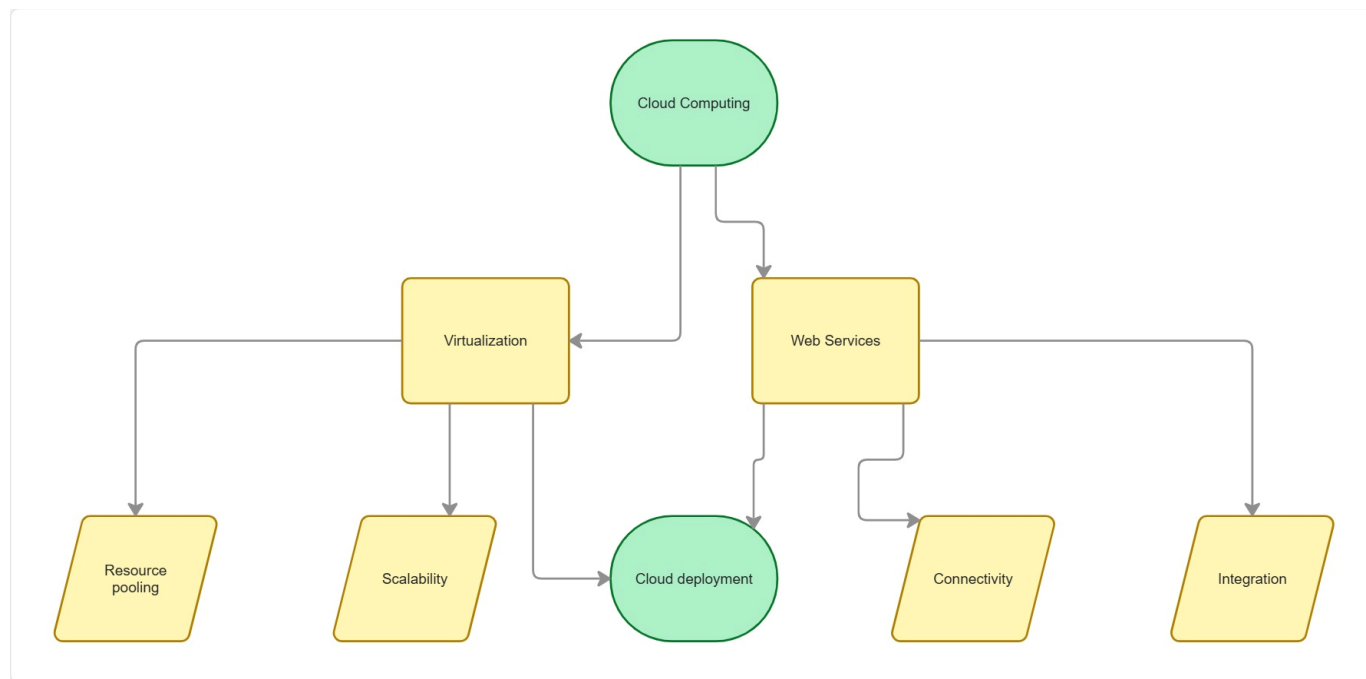
2. Evaluation of Cloud Computing



3. Service Models Map



4. Virtualization and Web Services



5. Answers:

1)

Hierarchical organization: Starting with Cloud Computing as the central node and branching into major topics makes the flow easy to understand.

- Categorization of service models (IaaS, PaaS, SaaS, XaaS): Clearly distinguishes different cloud offerings and provider-user responsibilities.
- Inclusion of enabling technologies: Virtualization and web services are shown as supporting technologies, demonstrating how cloud infrastructure operates.
- Relationship arrows: The connections illustrate dependencies between concepts instead of listing isolated terms.

2)

- Scalability allows the system to increase or decrease computing resources based on user demand.
- Elasticity automatically allocates and releases resources, reducing costs and improving performance.
- Resource management ensures efficient utilization of CPU, memory, storage, and network resources through virtualization.
- Without these features, the system would experience poor performance, higher costs, resource wastage, and reduced availability during peak workloads.

3)

- Web Services enable communication and data exchange between different cloud applications.
- APIs allow developers to integrate cloud services and automate application functionality.
- Cloud service models (IaaS, PaaS, SaaS, XaaS) provide different levels of infrastructure, platform, and software services based on user needs.
- Together, these technologies improve interoperability, simplify development, enhance flexibility, and increase overall system efficiency.

4)

- Virtualization and resource pooling improve hardware utilization and system scalability.
- Web services and APIs enable seamless connectivity and integration between applications.
- Cloud service models reduce management complexity by dividing responsibilities between providers and users.
- These interactions improve system reliability, deliver faster performance, and provide users with scalable and highly available cloud services.

5)

- Creating the concept map helped me understand the relationships between different cloud computing concepts.
- It improved my understanding of how virtualization, web services, and cloud service models work together.
- The visual representation made complex cloud architecture easier to learn and remember.
- If redesigned, I would include deployment models (Public, Private, Hybrid), security features, load balancing, auto-scaling, and examples of cloud providers like AWS, Azure, and Google Cloud.